

Roll No.

TPE-501

**B. TECH. PETROLEUM ENGINEERING (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

PETROLEUM GEOPHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How Geophysics plays a major role in petroleum industry ? What are the various earth properties are measured in Geophysics ? (CO1)
- (b) How is Geophysics used for imaging the subsurface structure ? What are the different recorded parameters used in this process ? (CO1)
- (c) What is passive and active method of geophysics. What is the objective of geophysical investigations ? (CO1)
2. (a) How many geophysical methods are available for oil exploration ? Explain the gravity and magnetic survey method in details. (CO2)
- (b) Explain Electromagnetic methods of geophysical survey ? Draw a neat diagram to explain it. Where is this survey useful ? (CO2)

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(2)

- (c) Explain electrical and well logging methods for oil exploration. What is the data format of Seismic data ? (CO3)
3. (a) What is seismic method for exploration ? What are the parameters measured in this method ? Why is seismic method widely accepted in exploration ? (CO3)
- (b) What is Geophones ? Explain its principle. How to avoid phenomenon of resonance while recording the data ? (CO3)
- (c) Explain Common Depth Point (CDP) Reflection Technique by drawing the figure. Write formula for foldage. (CO3)
4. (a) Explain seismic reflection prospecting methodology by drawing a figure. Write the formula for reflection coefficient. (CO4)
- (b) What are the various correction applied in reflection and refraction during the seismic data acquisition ? (CO4)
- (c) Explain NMO correction. What is diffraction ? Explain with figure. (CO4)
5. (a) What is Convolution model of seismic trace ? How is the seismic trace generated by this method ? (CO5)
- (b) Explain VSP. Why is it required ? What are the applications of VSP data ? (CO5)
- (c) Explain T-D relationship curve. Why is it important ? How is well to seismic tie performed ? (CO5)

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B. TECH. PETROLEUM ENGINEERING (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
WELL LOGGING AND FORMATION EVALUATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the role of logging tool and technology petroleum exploration and exploitation ? Explain in detail the various type of well logging services in E&P industry. (CO1)
- (b) With the help of a neat diagram, explain various subdivisions of invaded and un-invaded zones around a bore hole. Explain the reason of these divisions. (CO1)
- (c) Explain vertical resolution and depth of investigation of logging tools. Write these values for SP, GR, LLD, LLS, MSFL, RHOB, NPHI & DT. (CO1)
2. (a) Explain in detail for generation of spontaneous potential (SP) inside a bore hole. Why is the SP log and considered lithology identification log ? (CO2)

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(2)

- (b) Explain normal and reverse SP with diagram. What are the useful function of SP Log ? Explain in detail. (CO2)
- (c) How to calculate R_w from SP log ? Write the formula for volume of shale estimation from SP log. (CO2)
- 3. (a) What is the origin of GR log ? Write the operating principle of GR tool explain with figure. (CO3)
- (b) Why the SP log and gamma radiation curves (GR) are considered lithology curves and substitute for each other. What are the possible conditions when GR log can be helpful where SP log is non diagnostic ? (CO3)
- (c) Write formula for volume of shale estimation using GR Log. Calculate V_{shale} if $GR_{Log} = 50$, $GR_{shale} = 120$ & $GR_{clean} = 20$. (CO3)
- 4. (a) Drive the formula for porosity estimation from density log. Calculate porosity if $\rho_{matrix} = 2.65$, $\rho_{bulk} = 2.25$, $\rho_{fluid} = 1$ gm/cc. (CO4)
- (b) Explain principle of neutron logging. Draw figure of compensated neutron tool explaining the advantage over earlier neutron tools. (CO4)
- (c) Explain the principle of bore hole compensated sonic (BHC) tool with figure. Write Wyllie time average formula for estimation of porosity. Find out porosity of a formation, if $DT_{Log} = 90$, $DT_{Fluid} = 189$, $DT_{Matrix} = 55$ micro sec/fl. (CO4)
- 5. (a) Explain the working principle of OBMI2. How is this tool useful for imaging the geological features ? (CO5)
- (b) Write a short note on Platform Express 150, Decision Xpress Viewer V2.0, Sonic – MSIP Magnetic Resonance – MRX, Slim Cased Hole Formation Resistivity – SlimCHFR. (CO5)
- (c) Write principle of Dipole shear Imager tool with figure. What are its advantage over conventional sonic logging tools ? (CO5)

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B. TECH. (PE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
PETROLEUM PRODUCTION ENRINEERING-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is a Christmas tree ? Explain, with neat diagram the basic structure of Christmas tree and its use. (CO1)
- (b) What are packers and why they are used/run in petroleum industry. Explain group gathering station and its function. (CO1)
- (c) With help of a flow diagram explain oil facility and write all functions of an oil facility. (CO1)
2. (a) Illustrate principle of separators and for what purpose they are used in petroleum industry ? Construct and explain vertical three phase separator. (CO2)

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(2)

- (b) With the help of Amine treater or Sweater explain the process of Gas sweetening and also explain Glycol dehydration with the help of TEG. (CO2)
- (c) Describe Emulsion and all mechanisms involved in demulsification. (CO2)
3. (a) Describe Filling and pumping operations and Gas blanketing system and also write why these operations are used in Petroleum industry. (CO3)
- (b) Describe What are storage tanks, types of storage tanks and explain any two types of storage tanks. (CO3)
- (c) Describe what are the different methods by which we can control liquid leaks from tank and how leaks will be identified. (CO3)
4. (a) Explain Compressor and with the help of a flow chart write types of compressors. Also describe working principle and applications of centrifugal compressor. (CO4)
- (b) Write a short note on orifice meters and write 3-3 advantages and disadvantages of orifice meter, turbine meter and Coriolis meter. (CO4)
- (c) Write a short note on tank battery and tank breathing. (CO4)
5. (a) What do you understand by water injection and what are the sources of injected water ? (CO1, 5, 6)
- (b) Describe the Reciprocating Compressor, its working principle and applications. (CO1, 5, 6)
- (c) Write a short note on how produced water is treated in Petroleum Industry. (CO1, 5, 6)

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B. TECH. (PE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

RESERVOIR ENGINEERING II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Using a well is tested by producing it at a constant rate of 1500 stb/d for a period of 100 hours. It is suspected, from seismic and geological evidence, that the well is draining an isolated reservoir block which has approximately a 2 : 1 rectangular geometrical shape and the extended drawdown test is intended to confirm this. The reservoir data and flowing bottom hole pressures recorded during the test are detailed below and in table. (CO1, 2)

$h = 35$ ft, $c = 13 \times 10^{-6}$ /psi, Porosity = .30, $B_o = 1.55$ rb/stb, $r_w = .53$ ft,
Viscosity = 1 CP

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Flowing time (hours)	$P_{w.f}$ (psia)	Flowing time (hours)	$P_{w.f}$ (psia)
0	3500 (p_i)	20	2762
1	2917	30	2703
2	2900	40	2650
3	2888	50	2597
4	2879	60	2545
5	2869	70	2495
7.5	2848	80	2443
10	2830	90	2392
15	2794	100	2341

- (i) Calculate the effective permeability of well.
- (ii) Calculate skin factor of the well.
- (b) Illustrate dimensionless variables of the constant terminal rate solution of the radial diffusivity equation for transient state condition in terms of dimensionless pressure function. (CO1, 2)
- (c) Illustrate basic equation of radial flow of a single-phase fluid and basic differential equation for radial fluid flow. (CO1, 2)
2. (a) Explain the concept of constant terminal rate solution for transient state. (CO1, 2)
- (b) Explain t_D , in field units with the real time in hours and days and using dimensional analysis, show that both t_D and p_D are dimensionless. (CO1, 2)

- (c) With the help of a graph explain the concept of constant terminal rate solution for semi steady state. (CO1, 2)
3. (a) Write a short note on Water influx by describing water influx, occurrences and causes. (CO3)
- (b) Describe gas condensate and saturated oil reservoirs. (CO3)
- (c) (i) Calculate the water influx rate e_w in a reservoir whose pressure is stabilized at 3000 psi. Given :
- Initial reservoir pressure = 3500 psi
 - $dN_p/dt = 29,000$ STB/day
 - $B_o = 1.3$ bbl/STB
 - $GOR = 900$ scf/STB
 - $R_s = 700$ scf/STB
 - $B_g = 0.00082$ bbl/scf
 - $dW_p/dt = 0$
 - $B_{wo} = 1.0$ bbl/STB
- (ii) Calculate the cumulative water influx that results from a pressure drop of 200 psi at the oil-water contact with an encroachment angle of 80° . The reservoir-aquifer system is characterized by the following properties : (CO3)

	Reservoir	Aquifer
Radius. ft.	2600	10.000
Porosity	0.18	0.12
$C_w \text{psi}^{-1}$	4×10^{-6}	3×10^{-6}
$C_w \text{psi}^{-1}$	5×10^{-6}	4×10^{-6}
h .ft	20	25

4. (a) Write 9 typical measures of pressure in petroleum reservoirs. (CO3, 4)
- (b) Explain reservoir pressure measurement and reservoir management process. (CO3, 4)
- (c) Describe in brief oil reservoir performance prediction with all its phases. (CO3, 4)
5. (a) Define a gas reservoir has the following characteristics :
- $A = 3000$ acres
 - $H = 30$ ft
 - $\phi = 0.15$
 - $S_{wi} = 20\%$
 - $T = 150^\circ\text{F}$
 - $P_i = 2600$ psi

p	z
2600	0.82
1000	0.88
400	0.92

Calculate cumulative gas production and recovery factor at 1000 and 400 psi.

(CO5, 6)

(5)

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(b) The pressure history of a water-drive oil reservoir is given below :

t , days	p , days
0	3500 (p_i)
100	3450
200	3410
300	3380
400	3340

The aquifer is under a steadystate flowing condition with an estimated water influx constant of 130 bbl/day/psi. Calculate the cumulative water influx after 100, 200, 300 and 400 days using the steady-state model.

(CO5, 6)

(c) What is general theory of well testing ? Explain.

(CO5, 6)

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**B. TECH. PETROLEUM ENGINEERING (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

PETROLEUM REFINING ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the refineries feedstocks ? Classify the different physical and chemical processes involved in the refining of crude oil refining.

(CO1, CO2)

- (b) Define the following :

(CO1, CO2)

(i) Aniline point

(ii) Cloud point

(iii) Electric desalting

(iv) Pour point

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- (v) Fire point
 - (vi) Octane number
 - (vii) Cetane number
 - (viii) Viscosity index
- (c) Draw the flow sheet diagram of Atmospheric distillation unit (ADU) and discuss its process description with stepwise. (CO3)
2. (a) Petroleum refining processes and operations can be separated into five basic areas. Discuss each of them. (CO3, CO4)
- (b) Tabulate the physical and chemical process involved in petroleum refining. Explain the types of trays and plates in the distillation column. (CO1, CO4)
- (c) Explain solvent extraction ? What are the widely used extraction solvents ? What are the purpose of solvent extraction ? Design the flow sheet of aromatic solvent extraction unit. (CO1, CO5)
3. (a) What do you mean by thermal process in the refining of crude oil ? Discuss Visc breaking with the flow sheet diagram and illustrate the process description. (CO1)
- (b) Why we use delayed coking ? What are the *three* typical types of coke obtained ? Explain the steps only during the delayed coking. (CO1, CO2)
- (c) What are the different catalytic process ? How is cracking catalyzed ? What are the other reactions during the catalytic cracking ? Explain *one* of them. (CO2, CO5)

(3)

4. (a) What is FCC ? Design the flow sheet diagram of detailed FCC unit and explain their process description. (CO1, CO5)
- (b) Define the process hydro treating. Why do we do hydro treating ? Illustrate catalytic hydrodesulphurization in details with the help of trickle bed reactor diagram. (CO4, CO5)
- (c) What is hydrocracking ? Draw the different hydrocracking configuration. Explain the process description of hydrocracking. (CO1, CO6)
5. (a) Define reformates ? Draw and explain the process description of the flow sheet scheme of catalytic reforming. (CO4, CO6)
- (b) What do you mean by alkylation ? Explain the sulfuric acid alkylation with the suitable flowsheet diagram. (CO4, CO5)
- (c) Explain fluid coking and flexicoking. What are the different reactors used for catalytic hydrogenation ? Explain them with the labeled diagram. (CO2, CO3)

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B. TECH. (PE) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

MACHINE LEARNING FOR RESERVOIR CHARACTERIZATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) In detail explain the relationship between Artificial Intelligence, Machine Learning and Deep Learning. (CO1, CO2)
- (b) Describe perspectives and issues in machine learning with its types. Also explain differences between classification and regression with an example. (CO1, CO2)
- (c) Explain different parts of Measure of Central tendency and Measure of Variability. Give example for each part. (CO1, CO2)

P. T. O.

2. (a) What is Regression ? What is the difference between Dependent and Independent Variable ? Suppose you have the following data with one real-value input variable and one real-value output variable. What is leave-one out cross validation mean square error in case of linear regression ($Y = bX + c$) ? (CO1, CO4, CO5)

X (independent variable)	Y (dependent variable)
0	2
2	2
3	1

- (b) How are Advanced Ensemble Techniques different to Simple Ensemble Techniques ? How is Random forest related to Decision Tree ? How does Adaboost algorithm work ? (CO1, CO4, CO5)

- (c) Cluster the following eight points (with (x, y) representing locations) into three clusters :

A1 (2, 10), A2 (2, 5), A3 (8, 4), A4 (5, 8), A5 (7, 5), A6 (6, 4), A7 (1, 2), A8 (4, 9)

Initial cluster centers are : A1 (2, 10), A4 (5, 8) and A7 (1, 2).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is define as :

$$P(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Use K-Means Algorithm to find the three cluster centers after the second iteration. (CO1, CO4, CO5)

3. (a) Why is data preprocessed ? Explain Data Cleaning, Transformation and Reduction. (CO2, CO3)
- (b) What are some real-world applications of Time-Series Forecasting ? How the IQR (Interquartile Range) is used in Time Series Forecasting? How do you Normalise Time-Series Data ? (CO2, CO3)
- (c) What is Anomaly Detection ? What are different types of distance matrices in KNN Alorithm ? Perform KNN Classification Algorithm on the following dataset and predict the class for $X = 3$ and $Y = 7$ here $K = 3$. (CO2, CO3)

X	Y	Class
7	7	False
7	4	False
3	4	True
1	4	True

4. (a) Determine the range, variance and standard deviation of a sample from raw scores. A sample data of students scored the following : (CO4, CO5)

3, 45, 4, 41, 14, 75, 40, 11, 9, 5, 21, 61, 48, 61, 29, 61, 29, 51, 44, 67, 39, 23, 45, 57, 60, 63, 62

For the given data set, answer the following :

- What is the range of scores ?
- What is the variance in score ?
- What is the standard deviation in scores ?

- (b) What are generalized linear models ? State the differences between Linear Regression and Logistic Regression with examples and neat diagram. (CO4, CO5)
- (c) What are the different dimensionality reduction techniques ? Is dimensionality reduction a supervised or unsupervised machine learning technique ? Explain the following briefly : (CO4, CO5)
- (i) Principle Component Analysis
 - (ii) Random Projections
 - (iii) Feature Agglomeration
5. (a) Explain any *four* with example : (CO4, CO5)
- (i) Hierarchical clustering
 - (ii) Centroids-based clustering (Partitioning methods)
 - (iii) Distribution-based clustering
 - (iv) Density based Clustering (Model-based methods)
 - (v) Fuzzy Clustering
 - (vi) Constraint-based (Supervised Clustering)
- (b) In Reservoir Management elaborate Data Acquisition by describing the following points : (CO4, CO5)
- (i) Seismic Surveys
 - (ii) Well Logs
 - (iii) Conventional and special core analysis
 - (iv) Fluid Analyses

(5)

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(v) Static and flowing pressure measurements

(vi) Pressure-transient tests

(vii) Periodic well production tests

Records of monthly produced volumes of fluids (oil, gas and water) and monthly injected IOR/EOR volumes of fluids (water, gas, CO₂, steam, chemicals).

(c) Elaborate the process of Reservoir characterization by explaining the qualitative and quantitative assessment of data gathered from different sampling and preparation techniques, geographic selection through predictive analysis, conducting well test determined flow capacity, well bore condition, wettability and permeability for understanding reservoir performance. Draft a correct flow of above given factors with the help of Machine Learning Models required in each subprocess. (CO4, CO5)

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